Lecture 4

The Killing form and semisimple structure

Objectives. *Taking traces in a representation gives an invariant bilinear form on a Lie algebra. For the adjoint representation this is the Killing form. We prove that it detects semisimplicity and splits a semisimple algebra into simple ideals.*

By the end of this lecture you will be able to:

- Prove that trace forms are invariant and that orthogonal complements of ideals are ideals.
- State Cartan's criterion and explain how Jordan decomposition enters its proof.
- Prove that $\mathfrak{g}$ is semisimple exactly when its Killing form is nondegenerate.
- Decompose a semisimple algebra into simple ideals and compute the Killing form of $\mathfrak{sl}_2$.

Throughout, Lie algebras and modules are finite-dimensional over $\mathbb{C}$. The radical $\operatorname{Rad}\mathfrak{g}$ is the largest solvable ideal of $\mathfrak{g}$, as in Lecture 3. Results are marked [P] proved, [S] sketched or [C] cited.

4.1 Invariant trace forms

Definition 4.1 (Invariant form)**.** A bilinear form $\beta : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$ is *invariant* if

$$\beta([x,y],z) = \beta(x,[y,z]) \quad \text{for all } x,y,z \in \mathfrak{g}.$$

For a subspace $\mathfrak{a} \subseteq \mathfrak{g}$, put

$$\mathfrak{a}^{\perp} = \{x \in \mathfrak{g} : \beta(x,a) = 0 \text{ for all } a \in \mathfrak{a}\}.$$

The *radical* of β is $\mathfrak{g}^{\perp}$. The form is *nondegenerate* if its radical is zero.

All forms below are symmetric. So it does not matter on which side orthogonality is tested. Do not confuse the radical $\mathfrak{g}^{\perp}$ of a form with the radical $\operatorname{Rad}\mathfrak{g}$ of the algebra. Example 4.12 shows that they can differ.

Definition 4.2 (Trace form and Killing form)**.** Let $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation. Its *trace form* is

$$\beta_\rho(x,y) = \operatorname{tr}\big(\rho(x)\rho(y)\big).$$

The *Killing form* of $\mathfrak{g}$ is the trace form of the adjoint representation:

$$\kappa_{\mathfrak{g}}(x,y) = \operatorname{tr}(\operatorname{ad} x \operatorname{ad} y).$$

We write κ when the algebra is clear. The Killing form depends only on the Lie algebra.

Proposition 4.3 (Invariance [P])**.** *Every trace form is symmetric and invariant. In particular, so is* $\kappa_{\mathfrak{g}}$.

Proof. Symmetry of β_ρ follows from $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. For $A, B, C \in \operatorname{End}(V)$, cyclicity of the trace gives $\operatorname{tr}(ACB) = \operatorname{tr}(BAC)$. Expanding the commutators and using this equality gives

$$\begin{aligned}\operatorname{tr}([A,B]C) &= \operatorname{tr}(ABC) - \operatorname{tr}(BAC),\\ \operatorname{tr}(A[B,C]) &= \operatorname{tr}(ABC) - \operatorname{tr}(ACB) = \operatorname{tr}(ABC) - \operatorname{tr}(BAC).\end{aligned}$$

Now put $A = \rho(x)$, $B = \rho(y)$ and $C = \rho(z)$. Since ρ is a representation, $\rho([x,y]) = [A,B]$ and $\rho([y,z]) = [B,C]$. Hence

$$\beta_\rho([x,y],z) = \operatorname{tr}([A,B]C) = \operatorname{tr}(A[B,C]) = \beta_\rho(x,[y,z]).$$

■

Lemma 4.4 (Orthogonal ideals [P])**.** *Let β be a symmetric invariant form on $\mathfrak{g}$, and let $\mathfrak{a}$ be an ideal. Then $\mathfrak{a}^\perp$ is an ideal. In particular, the radical of β is an ideal.*

Proof. Let $x \in \mathfrak{a}^\perp$, $y \in \mathfrak{g}$ and $a \in \mathfrak{a}$. Antisymmetry of the bracket and invariance of β give

$$\beta([y,x],a) = -\beta([x,y],a) = -\beta(x,[y,a]) = 0,$$

since $[y,a] \in \mathfrak{a}$ and $x \in \mathfrak{a}^\perp$. As $a \in \mathfrak{a}$ was arbitrary, $[y,x] \in \mathfrak{a}^\perp$. The radical $\mathfrak{g}^\perp$ is the case $\mathfrak{a} = \mathfrak{g}$. ■

Lemma 4.5 (Killing form of an ideal [P])**.** *If $\mathfrak{a}$ is an ideal of $\mathfrak{g}$, then $\kappa_{\mathfrak{a}}(x,y) = \kappa_{\mathfrak{g}}(x,y)$ for all $x, y \in \mathfrak{a}$.*

Proof. First, let W be a subspace of a vector space V, and let $\varphi \in \operatorname{End}(V)$ satisfy $\varphi(V) \subseteq W$. Extend a basis of W to a basis of V. In this basis the matrix of φ is

$$\begin{pmatrix} A & * \\ 0 & 0 \end{pmatrix},$$

where A is the matrix of $\varphi|_W$. Hence $\operatorname{tr}\varphi = \operatorname{tr}(\varphi|_W)$.

Now let $x, y \in \mathfrak{a}$ and put $\varphi = \operatorname{ad} x \operatorname{ad} y \in \operatorname{End}(\mathfrak{g})$. Since $\mathfrak{a}$ is an ideal, $\operatorname{ad} x$ maps $\mathfrak{g}$ into $\mathfrak{a}$, so $\varphi(\mathfrak{g}) \subseteq \mathfrak{a}$. On $\mathfrak{a}$, the map φ equals $\operatorname{ad}_{\mathfrak{a}} x \operatorname{ad}_{\mathfrak{a}} y$. The first paragraph, with $V = \mathfrak{g}$ and $W = \mathfrak{a}$, gives

$$\kappa_{\mathfrak{g}}(x,y) = \operatorname{tr}\varphi = \operatorname{tr}(\varphi|_{\mathfrak{a}}) = \kappa_{\mathfrak{a}}(x,y).$$

■

The ideal hypothesis matters. The line $\mathbb{C}h \subseteq \mathfrak{sl}_2$ is abelian, so $\kappa_{\mathbb{C}h}(h,h) = 0$. This subalgebra is not an ideal. § 4.4 shows that $\kappa_{\mathfrak{sl}_2}(h,h) = 8$.

4.2 Cartan's criterion

Recall the linear Jordan decomposition from Lecture 3. Each $x \in \mathrm{End}(V)$ has a unique decomposition

$$x = x_s + x_n, \qquad x_s x_n = x_n x_s,$$

with x_s diagonalisable and x_n nilpotent. Both parts are polynomials in x without constant term.

Theorem 4.6 (Cartan's criterion [C]). *Let V be a finite-dimensional complex vector space and $L \subseteq \mathfrak{gl}(V)$ a Lie subalgebra. Suppose*

$$\mathrm{tr}(xy) = 0 \quad \text{for all } x \in [L, L],\ y \in L.$$

Then L is solvable. (Humphreys, §4.3, Theorem; Erdmann–Wildon, Chapter 9.)

Sketch of proof [S]. It suffices to show that every $x \in [L, L]$ is nilpotent. Engel's theorem (Lecture 2) then makes $[L, L]$ nilpotent, so L is solvable.

Put $M = \{z \in \mathfrak{gl}(V) : [z, L] \subseteq [L, L]\}$, so $L \subseteq M$. For $a, b \in L$ and $z \in M$, invariance of the trace form gives

$$\mathrm{tr}([a, b]z) = \mathrm{tr}(a[b, z]) = 0,$$

because $[b, z] \in [L, L]$ and $a \in L$. The brackets $[a, b]$ span $[L, L]$. Hence $\mathrm{tr}(xz) = 0$ for all $x \in [L, L]$ and $z \in M$.

Since $[L, L] \subseteq L \subseteq M$, the next step applies to every $x \in [L, L]$.

The delicate step is this: if $x \in M$ and $\mathrm{tr}(xz) = 0$ for all $z \in M$, then x is nilpotent.

Fix an eigenbasis of $s = x_s$, and let $a_1, \ldots, a_m$ be the eigenvalues of s, listed with multiplicity, where $m = \dim V$.

Let $E \subseteq \mathbb{C}$ be the $\mathbb{Q}$-span of the a_i, and let $f : E \to \mathbb{Q}$ be $\mathbb{Q}$-linear. Let y be diagonal in the same basis, with entries $f(a_i)$.

By Lagrange interpolation, y is a polynomial in s, so y commutes with x_n and $x_n y$ is nilpotent. Hence $\mathrm{tr}(x_n y) = 0$. Assume $y \in M$. Since $x = s + x_n$ and sy is diagonal with entries $a_i f(a_i)$, the hypothesis on x gives

$$0 = \mathrm{tr}(xy) = \mathrm{tr}(sy) + \mathrm{tr}(x_n y) = \sum_{i=1}^{m} a_i f(a_i).$$

Applying the $\mathbb{Q}$-linear map f gives $\sum_i f(a_i)^2 = 0$ in $\mathbb{Q}$. So every $f(a_i) = 0$. The a_i span E, so $f = 0$.

Since f was an arbitrary $\mathbb{Q}$-linear functional, $E^* = 0$. Hence $E = 0$ and $s = 0$, so $x = x_n$ is nilpotent.

The omitted step is $y \in M$. Humphreys shows that $\mathrm{ad}\, y$ is a polynomial in $\mathrm{ad}\, x$ without constant term. Since $x \in M$, the map $\mathrm{ad}\, x$ sends L into $[L, L]$ and preserves $[L, L]$. So that polynomial sends L into $[L, L]$, which gives $y \in M$.

The polynomial construction uses Lagrange interpolation on the differences $a_i - a_j$ and the fact that $\mathrm{ad}\, s$ is the semisimple part of $\mathrm{ad}\, x$ (Humphreys, §4.2, Proposition and Lemma A). The rational functional f replaces an ordering, which $\mathbb{C}$ does not have.

Lemma 4.7 (Killing-null subalgebras [P]). *Let $\mathfrak{a}$ be a subalgebra of $\mathfrak{g}$ with $\kappa_{\mathfrak{g}}(x, y) = 0$ for all $x, y \in \mathfrak{a}$. Then $\mathfrak{a}$ is solvable.*

Proof. The image $L = \operatorname{ad}(\mathfrak{a})$ is a subalgebra of $\mathfrak{gl}(\mathfrak{g})$. For $x, y \in \mathfrak{a}$,

$$\operatorname{tr}(\operatorname{ad} x \operatorname{ad} y) = \kappa_{\mathfrak{g}}(x, y) = 0.$$

Since $[L, L] \subseteq L$, Theorem 4.6 applies, and L is solvable. The restriction of ad to $\mathfrak{a}$ has kernel $\mathfrak{a} \cap Z(\mathfrak{g})$, an abelian ideal of $\mathfrak{a}$. The quotient is isomorphic to L. By Lecture 3, an algebra with a solvable ideal and solvable quotient is solvable. Hence $\mathfrak{a}$ is solvable. ∎

4.3 The Killing criterion

Definition 4.8 (Semisimple and simple)**.** A Lie algebra $\mathfrak{g}$ is *semisimple* if $\operatorname{Rad}\mathfrak{g} = 0$, that is, if $\mathfrak{g}$ has no nonzero solvable ideal. It is *simple* if it is nonabelian and its only ideals are 0 and $\mathfrak{g}$.

The zero algebra is semisimple under this convention. A simple algebra is semisimple. Indeed, $[\mathfrak{g}, \mathfrak{g}]$ is an ideal, and it is nonzero because $\mathfrak{g}$ is nonabelian. So $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. The derived series is constant and nonzero, so $\mathfrak{g}$ is not solvable. The ideal $\operatorname{Rad}\mathfrak{g}$ is therefore not $\mathfrak{g}$, so it is zero.

Lemma 4.9 (Abelian ideals [P])**.** *If $\mathfrak{g}$ has a nonzero solvable ideal $\mathfrak{s}$, then $\mathfrak{g}$ has a nonzero abelian ideal.*

Proof. If J is an ideal of $\mathfrak{g}$, so is $[J, J]$. For $x \in \mathfrak{g}$ and $a, b \in J$, the Jacobi identity gives

$$[x, [a, b]] = [[x, a], b] + [a, [x, b]] \in [J, J],$$

since $[x, a]$ and $[x, b]$ lie in J. By induction, every derived term $\mathfrak{s}^{(k)}$ is an ideal of $\mathfrak{g}$. The last nonzero term is abelian. ∎

Theorem 4.10 (Killing criterion [P])**.** *A Lie algebra $\mathfrak{g}$ is semisimple if and only if $\kappa_{\mathfrak{g}}$ is nondegenerate.*

Proof. Let $S = \mathfrak{g}^{\perp}$ be the radical of $\kappa = \kappa_{\mathfrak{g}}$. By Lemma 4.4, S is an ideal. Since $S = \mathfrak{g}^{\perp}$, the form κ vanishes on $S \times S$, so S is solvable by Lemma 4.7. If $\mathfrak{g}$ is semisimple, then $S \subseteq \operatorname{Rad}\mathfrak{g} = 0$.

Conversely, suppose $S = 0$. Let $\mathfrak{a}$ be an abelian ideal. Take $x \in \mathfrak{a}$ and $y \in \mathfrak{g}$, and put $\varphi = \operatorname{ad} x \operatorname{ad} y$. For $z \in \mathfrak{g}$,

$$\varphi(z) = [x, [y, z]] \in \mathfrak{a},$$
$$\varphi^2(z) = [x, [y, \varphi(z)]] \in [\mathfrak{a}, \mathfrak{a}] = 0.$$

Thus φ is nilpotent and $\kappa(x, y) = \operatorname{tr}\varphi = 0$. Since $y \in \mathfrak{g}$ was arbitrary, $x \in S$. Hence $\mathfrak{a} \subseteq S = 0$. By Lemma 4.9, $\mathfrak{g}$ has no nonzero solvable ideal, so $\mathfrak{g}$ is semisimple. ∎

4.4 Two Killing forms

Example 4.11 (The Killing form of $\mathfrak{sl}_2$)**.** Use the ordered basis (h, e, f), with brackets

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h.$$

Column j of $D(\operatorname{ad} x)$ holds the coordinates of $[x, b_j]$, where b_j is the jth basis vector. For instance, $[e,h] = -2e$ and $[e,f] = h$. This gives

$$D(\operatorname{ad} h) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \qquad D(\operatorname{ad} e) = \begin{pmatrix} 0 & 0 & 1 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad D(\operatorname{ad} f) = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

Two trace computations suffice:

$$D(\operatorname{ad} h)^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad D(\operatorname{ad} e)D(\operatorname{ad} f) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence

$$\kappa(h,h) = 8 \quad \text{and} \quad \kappa(e,f) = 4.$$

The products $D(\operatorname{ad} h)D(\operatorname{ad} e)$ and $D(\operatorname{ad} h)D(\operatorname{ad} f)$ each have one nonzero entry, equal to -4, off the diagonal. So $\kappa(h,e) = \kappa(h,f) = 0$. The operators $\operatorname{ad} e$ and $\operatorname{ad} f$ are nilpotent, so $\kappa(e,e) = \kappa(f,f) = 0$. The Gram matrix is

$$G = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}, \qquad \det G = -128 \neq 0.$$

By Theorem 4.10, $\mathfrak{sl}_2$ is semisimple. For $X, Y \in \mathfrak{sl}_n(\mathbb{C})$ and $n \geq 2$, Exercise 4.1 proves

$$\kappa_{\mathfrak{sl}_n}(X,Y) = 2n \operatorname{tr}(XY).$$

Example 4.12 (A degenerate Killing form)**.** Let $\mathfrak{b} \subseteq \mathfrak{gl}_2(\mathbb{C})$ be the upper triangular algebra, with ordered basis (E_{11}, E_{12}, E_{22}). For

$$X = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix},$$

direct multiplication gives

$$[X, E_{11}] = -bE_{12}, \qquad [X, E_{12}] = (a-d)E_{12}, \qquad [X, E_{22}] = bE_{12}.$$

Hence

$$D(\operatorname{ad} X) = \begin{pmatrix} 0 & 0 & 0 \\ -b & a-d & b \\ 0 & 0 & 0 \end{pmatrix}.$$

Let Y have entries u, v, w in the positions of a, b, d. Only the middle row of $D(\operatorname{ad} X)D(\operatorname{ad} Y)$ is nonzero, and its middle entry is $(a-d)(u-w)$. Therefore

$$\kappa_{\mathfrak{b}}(X,Y) = (a-d)(u-w).$$

The radical of $\kappa_{\mathfrak{b}}$ is $\{X : a = d\} = \mathbb{C}E_{12} \oplus \mathbb{C}I_2$. It is a nonzero abelian ideal, so $\mathfrak{b}$ is not semisimple. This agrees with the solvability of $\mathfrak{b}$ from Lecture 3. The radical of $\kappa_{\mathfrak{b}}$ is smaller than $\operatorname{Rad}\mathfrak{b} = \mathfrak{b}$.

Nondegenerate is not positive definite

Nondegeneracy says that the map

$$\mathfrak{g} \longrightarrow \mathfrak{g}^*, \qquad x \longmapsto \kappa(x, \cdot)$$

is injective. It does not say that $\kappa(x, x) \neq 0$ for every nonzero x. For example, in $\mathfrak{sl}_2$,

$$\kappa(e, e) = 0, \qquad \kappa(e, f) = 4.$$

A symmetric bilinear form β on a nonzero complex space cannot be positive on all nonzero vectors, since

$$\beta(ix, ix) = -\beta(x, x).$$

Exercise 4.5 compares two real restrictions. Lecture 8 uses a positive definite restriction.

4.5 Decomposition into simple ideals

Theorem 4.13 (Simple ideals [P]). *Let $\mathfrak{g}$ be semisimple. There are simple ideals $\mathfrak{g}_1, \dots, \mathfrak{g}_t$ of $\mathfrak{g}$ with*

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_t.$$

Every simple ideal of $\mathfrak{g}$ is one of the $\mathfrak{g}_i$. The Killing form of $\mathfrak{g}_i$ is the restriction of $\kappa_{\mathfrak{g}}$.

Proof. Orthogonal complements. Let $\mathfrak{a}$ be an ideal, and take $\mathfrak{a}^\perp$ with respect to $\kappa = \kappa_{\mathfrak{g}}$. By Lemma 4.4, $\mathfrak{a}^\perp$ is an ideal, hence so is $\mathfrak{a} \cap \mathfrak{a}^\perp$. The form κ vanishes on this intersection, so it is solvable by Lemma 4.7. Semisimplicity gives $\mathfrak{a} \cap \mathfrak{a}^\perp = 0$. The map $\mathfrak{g} \to \mathfrak{a}^*$, $x \mapsto \kappa(x, \cdot)|_{\mathfrak{a}}$, has kernel $\mathfrak{a}^\perp$. By rank–nullity, $\dim \mathfrak{a}^\perp \geq \dim \mathfrak{g} - \dim \mathfrak{a}$. Therefore, since $\mathfrak{a}$ and $\mathfrak{a}^\perp$ are ideals,

$$\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp, \qquad [\mathfrak{a}, \mathfrak{a}^\perp] \subseteq \mathfrak{a} \cap \mathfrak{a}^\perp = 0.$$

Ideals of a summand. Let $\mathfrak{c}$ be an ideal of $\mathfrak{a}$. Then

$$[\mathfrak{g}, \mathfrak{c}] = [\mathfrak{a}, \mathfrak{c}] + [\mathfrak{a}^\perp, \mathfrak{c}] = [\mathfrak{a}, \mathfrak{c}] \subseteq \mathfrak{c},$$

so $\mathfrak{c}$ is an ideal of $\mathfrak{g}$. The same holds for ideals of $\mathfrak{a}^\perp$. A solvable ideal of $\mathfrak{a}$ or of $\mathfrak{a}^\perp$ is therefore a solvable ideal of $\mathfrak{g}$, hence zero. Thus $\mathfrak{a}$ and $\mathfrak{a}^\perp$ are semisimple.

Existence. Induct on $\dim \mathfrak{g}$. If $\mathfrak{g} = 0$, take $t = 0$. Otherwise choose a nonzero ideal $\mathfrak{a}$ of smallest dimension. Its ideals are ideals of $\mathfrak{g}$. By the minimality of $\dim \mathfrak{a}$, they are 0 and $\mathfrak{a}$. It is nonzero and semisimple, hence nonabelian. So $\mathfrak{a}$ is simple. The ideal $\mathfrak{a}^\perp$ is semisimple of smaller dimension. By induction it is a direct sum of simple ideals of $\mathfrak{a}^\perp$, and these are ideals of $\mathfrak{g}$. Together with $\mathfrak{a}$, they give the required decomposition of $\mathfrak{g}$.

Uniqueness. Let $\mathfrak{c}$ be a simple ideal of $\mathfrak{g}$. The centre $Z(\mathfrak{g})$ is an abelian ideal, so it is zero. Since $\mathfrak{c} \neq 0$, it is not central, so $[\mathfrak{g}, \mathfrak{c}] \neq 0$. This bracket is an ideal of $\mathfrak{g}$ inside $\mathfrak{c}$, hence an ideal of the simple algebra $\mathfrak{c}$. So $[\mathfrak{g}, \mathfrak{c}] = \mathfrak{c}$. Now

$$\mathfrak{c} = [\mathfrak{g}, \mathfrak{c}] = \sum_{i=1}^{t} [\mathfrak{g}_i, \mathfrak{c}], \qquad [\mathfrak{g}_i, \mathfrak{c}] \subseteq \mathfrak{g}_i \cap \mathfrak{c}.$$

Choose i with $[\mathfrak{g}_i, \mathfrak{c}] \neq 0$. Then $\mathfrak{g}_i \cap \mathfrak{c}$ is a nonzero ideal of $\mathfrak{g}$. It is an ideal of both simple algebras $\mathfrak{g}_i$ and $\mathfrak{c}$. Hence $\mathfrak{g}_i = \mathfrak{g}_i \cap \mathfrak{c} = \mathfrak{c}$.

The last assertion is Lemma 4.5. ∎

Corollary 4.14 (Ideals of a semisimple algebra [P]). *Let $\mathfrak{g}$ be semisimple, decomposed as in Theorem 4.13.*

(i) $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$.

(ii) Every ideal of $\mathfrak{g}$ is the sum of some of the $\mathfrak{g}_i$.

Proof. (i) Each $\mathfrak{g}_i$ is simple, so $[\mathfrak{g}_i,\mathfrak{g}_i] = \mathfrak{g}_i$. Hence $[\mathfrak{g},\mathfrak{g}] \supseteq \sum_i [\mathfrak{g}_i,\mathfrak{g}_i] = \mathfrak{g}$.

(ii) Let $\mathfrak{a}$ be an ideal. The proof of Theorem 4.13 shows that $\mathfrak{a}$ is semisimple and that its ideals are ideals of $\mathfrak{g}$. By Theorem 4.13 applied to $\mathfrak{a}$, the ideal $\mathfrak{a}$ is a direct sum of simple ideals of $\mathfrak{a}$. These are simple ideals of $\mathfrak{g}$, hence among the $\mathfrak{g}_i$. ∎

4.6 Imports for root theory

Recall from Lecture 1 that a *derivation* of $\mathfrak{g}$ is a linear map $D\colon \mathfrak{g} \to \mathfrak{g}$ with

$$D[y,z] = [Dy,z] + [y,Dz] \quad \text{for all } y,z \in \mathfrak{g}.$$

The derivations form a Lie subalgebra $\operatorname{Der}\mathfrak{g}$ of $\mathfrak{gl}(\mathfrak{g})$. For semisimple $\mathfrak{g}$, the centre is zero, so ad is injective.

Theorem 4.15 (Inner derivations [C]). *Let $\mathfrak{g}$ be semisimple. Then* $\operatorname{ad}\colon \mathfrak{g} \to \operatorname{Der}\mathfrak{g}$ *is an isomorphism of Lie algebras.* (Humphreys, §5.3, Theorem; Erdmann–Wildon, Chapter 9.)

Theorem 4.16 (Abstract Jordan decomposition [C]). *Let $\mathfrak{g}$ be semisimple and $x \in \mathfrak{g}$. There are unique $x_s, x_n \in \mathfrak{g}$ with*

$$x = x_s + x_n, \qquad [x_s, x_n] = 0,$$

such that $\operatorname{ad} x_s$ *is diagonalisable and* $\operatorname{ad} x_n$ *is nilpotent. Moreover,* $\operatorname{ad} x_s$ *and* $\operatorname{ad} x_n$ *are the semisimple and nilpotent parts of* $\operatorname{ad} x$. (Humphreys, §5.4.)

The second import uses the first. The semisimple and nilpotent parts of $\operatorname{ad} x$ are derivations of $\mathfrak{g}$ (Humphreys, §4.2, Lemma B). Hence Theorem 4.15 places them in $\operatorname{ad}\mathfrak{g}$. Semisimplicity is also needed for uniqueness.

One consequence is used in Lecture 7. If $[x,y] = 0$, then $[x_s,y] = 0$. Indeed, $\operatorname{ad} x_s = p(\operatorname{ad} x)$ for a polynomial p without constant term, and $(\operatorname{ad} x)(y) = 0$. Lecture 7 also imports the compatibility of this decomposition with representations.

Self-checks

(1) Is the Killing form of $\mathfrak{gl}_2(\mathbb{C})$ nondegenerate?

(2) Let $\mathfrak{g} = \mathfrak{sl}_2 \times \mathfrak{sl}_2$. List its ideals. Is the diagonal $\{(x,x)\}$ an ideal?

(3) Give a nonzero $x \in \mathfrak{sl}_2$ with $\kappa(x,x) = 0$. Why is this consistent with nondegeneracy?

Answers. (1) No. Since $\operatorname{ad} I_2 = 0$, the identity lies in the radical of the Killing form. This matches the abelian ideal $\mathbb{C}I_2$. (2) The algebra $\mathfrak{sl}_2$ is simple by Exercise 1.3. Each $\operatorname{ad}(x_1,x_2)$ preserves both factors, so $\kappa_{\mathfrak{g}}$ is the sum of two nondegenerate forms. Thus $\mathfrak{g}$ is semisimple. Its simple ideals are the two factors, so by Corollary 4.14 its ideals are 0, the two factors and $\mathfrak{g}$. The diagonal is not an ideal, since

$$[(y,0),(x,x)] = ([y,x],0).$$

This lies outside the diagonal whenever $[y,x] \neq 0$, for example for $y = h$ and $x = e$. (3) Take $x = e$. Nondegeneracy needs some y with $\kappa(e,y) \neq 0$, and $\kappa(e,f) = 4$.

Summary

Trace forms are invariant, so orthogonal complements of ideals are ideals. Cartan's criterion is cited with a proof sketch. It shows that a subalgebra on which the Killing form vanishes is solvable. Hence $\mathfrak{g}$ is semisimple exactly when $\kappa_{\mathfrak{g}}$ is nondegenerate.

A semisimple algebra is the direct sum of its simple ideals, and these are unique. Nondegeneracy is not positivity. For example, $e \neq 0$ in $\mathfrak{sl}_2$, but $\kappa(e, e) = 0$. Lecture 5 uses a nondegenerate trace form to build the Casimir operator.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 4.2–4.3 and 5.1–5.4. Erdmann–Wildon, *Introduction to Lie Algebras*, Chapter 9. Exercise Sheet 4 treats orthogonal ideals, quotients and the Killing form of $\mathfrak{sl}_n$.